HU Extension CSCI E-89 Deep Learning

Assignment 06

Handed out: 10/09/2026 Due by 1 PM, Saturday, 10/17/2026

You are provided with the Geron's notebook from chapter 18 of his PyTorch book, `18_autoencoders_gans_and_diffusion_models.ipynb`. Please create your own notebook only with your solutions to the following problems. Please do not submit modified version of Geron's notebook.

Problem 1. Plots of training data on slides 24 and 28 are actually from another example and not from the above notebook. Use the following code to generate a new set of 3D data `X_train`:

```
import numpy as np
np.random.seed(4)
def generate_3d_data(m, w1=0.1, w2=0.3, noise=0.1):
    angles = np.random.rand(m) * 3 * np.pi / 2 - 0.5
    data = np.empty((m, 3))
    data[:, 0] = np.cos(angles) + np.sin(angles)/2 + noise *
           np.random.randn(m) / 2
    data[:, 1] = np.sin(angles) * 0.7 + noise *
           np.random.randn(m) / 2
    data[:, 2] = data[:, 0] * w1 + data[:, 1] * w2 + noise *
           np.random.randn(m)
    return data

X_train = generate_3d_data(60)
X_train = X_train - X_train.mean(axis=0, keepdims=0)
```

Create validation data, train the autoencoder with code given on slide 25 and plot resulting 2D `codings`.
(20%)

Problem 2. Train a deep autoencoder with the architecture presented on slide 33 and train it on MNIST data (images of handwritten digits). Present the original images of the first 5 elements of that data set and their reconstructed images.
(10%)

Problem 3. Pass 4 images from the Fashion MNIST data set to the autoencoder trained in Problem 2 and present their reconstructed images. This is a demonstration of the anomaly detection. Calculate the loss value that such autoencoder produces for MNIST images and the loss values for Fashion MNIST images. Compare
(10%)

Problem 4. Please generate a 2D t-SNE display for MNIST data set similar to the display on slide 49.
(20%)

Problem 5. Implement a convolutional denoising autoencoder with dropout layer and train it with MNIST data set. As the dropout parameter use 0.20. Produce 5 original images of random digits and their reconstructed images. Change the dropout parameter to 0.10 and compare new reconstructed images with the previous one. Just produce a qualitative assessment of the change in quality of the reconstructed images if any.
(20%)

Problem 6. Train a Variational Autoencoder on MNIST dataset with coding dimension of 2. Pick the mean coding values for two random images of two digits, for example 3 and 7. Create a set of 20 values uniformly distributed between those two coding values. Generate reconstruction images for every one of those coding values and present them as a 5x4 matrix.
(20%)

Please add your name to the top of your notebooks.
We expect you to submit one notebook (*.ipynb file) along with its HTML images.